

BioRaman

Raman Hyperspectral Map Analysis for Biophysics — User Manual

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Audience: new and experienced users · Desktop GUI (Python / Tkinter)

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A desktop application for loading, visualising and analysing Raman hyperspectral maps — covering data import from many instrument formats, RAMANMETRIX-compatible dataset and metadata handling, reproducible preprocessing recipes, batch processing, univariate and ratio mapping, curve fitting, PCA, clustering, MCR-ALS, N-FINDR endmember extraction, peak identification, spectra comparison, quality-control maps, HTML reporting and 3D confocal rendering. A headless command-line mode is included for pipeline use.

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1 Introduction

BioRaman is a desktop application for working with Raman hyperspectral maps — datasets where a full Raman spectrum is recorded at every pixel of a 2-D (or 3-D) sample area. It reads a wide range of instrument and text formats and provides a complete workflow from raw spectra to publication-ready chemical maps and statistics.

The program is organised around a single main window showing the map on one side and the spectrum at the cursor on the other. All advanced analyses (PCA, clustering, unmixing, 3D rendering, and so on) open in their own dedicated windows from the menu bar, so you can keep several analyses open at once.

What is new in v0.10.0

- **Preprocessing recipes** — save and load every preprocessing parameter as a JSON file for reproducible, shareable analyses.
- **Non-destructive reprocessing** — the raw cube is kept in memory, so a changed recipe can be re-applied without re-reading the file.
- **Save Processed Data** — export the preprocessed cube to NPZ, HDF5, CSV, TXT or MAT.
- **Batch processing** — apply one recipe to a whole folder, with a summary CSV.
- **Headless command-line mode** — run batch preprocessing with no GUI.
- **Quality-control maps** — per-pixel SNR, intensity and saturation.
- **HTML analysis report** and **session save/restore**.
- **Cluster validation** via the mean silhouette score.

How to use this manual

Sections 2–11 cover the everyday workflow and are recommended reading for new users. Sections 12–19 describe the advanced analysis windows and assume some familiarity with chemometrics and spectroscopy, though each begins with a plain-language summary. Sections 20–24 cover the new export, batch, reporting and command-line features. Sections 25–28 and the appendices are reference material.

Note BioRaman follows semantic versioning. Any change that affects analytical output is flagged in the CHANGELOG and triggers at least a minor version bump.

2 Installation & Launching

BioRaman requires Python 3.9 or newer. Install the core and recommended dependencies with pip:

```
pip install numpy scipy matplotlib pybaselines renishawWiRE pillow
pip install scikit-learn pandas seaborn openpyxl
```

Optional packages enable additional features:

Package	Enables	Install
ramanspy	WITec .wip files	pip install ramanspy
h5py	HDF5 (.h5) export of processed data	pip install h5py
pandas + openpyxl	Reading XLS/XLSX spectra and metadata tables	pip install pandas openpyxl

Plain text spectra (.txt / .csv / .dpt / .dat / .jdx / .asc) need no extra packages — they are read by a built-in parser.

Note The 3D viewer uses `mpl_toolkits.mplot3d`, which ships with matplotlib and needs no separate installation.

Launch the application

Run the program from the folder that contains it:

```
python bioraman.py
```

To process files without the interface, see section 24 (Headless Command-Line Mode).

3 The Main Window at a Glance

When the application starts you see the main window, divided into four zones:

Zone	What it contains
Menu bar	File, View, Preprocessing, Analysis and Help menus — the entry point to every feature and analysis window.
Toolbar	Quick-access buttons (Open file, White Light, QC Maps, Batch, Save Map, Save Spec, Save Processed, and the analysis launchers) plus the MODE selector, the CMAP selector and a progress bar.
Map panel (left)	The colour-coded chemical map of the sample. You click here to pick a pixel, and draw ROIs here.
Spectrum panel (right)	The Raman spectrum at the selected pixel (or the mean spectrum of a drawn ROI). Right-click adds a spectrum to the comparison overlay.

The MODE selector controls what the main map shows: Ratio (a band-ratio or single-band intensity map), A+B RGB (two bands rendered as colour channels) and White Light (the optical reference image, if loaded).

The CMAP selector chooses the colour map (look-up table) used to render intensity maps.

4 Loading Data & Supported File Formats

Open a file with **File > Open Raman file** (Ctrl+O) or the toolbar button. The format is detected from the extension.

Format	Reader / backend	Notes
.wdf	renishawWiRE	Renishaw WiRE binary maps, line scans and single spectra. Embedded white-light image is loaded automatically when present.
.wip	RamanSPy (or wip_loader)	WITec Project binary. Requires pip install ramanspy. Single spectra / line scans are reshaped to a cube.
.txt .csv	built-in reader	ASCII spectra and tables (see 4.2). No extra packages needed.
.dpt .dat .asc .jdx	built-in reader	Generic two-column or multi-column ASCII exports.
.xlsx .xls	pandas + openpyxl	Tabular spectra (mainly PCA input).

4.2 Text / CSV layouts understood by the reader

The native text reader auto-detects the delimiter (tab, semicolon, comma or whitespace), skips comment lines beginning with # or %, drops a non-numeric header row, and copes with European decimal commas. It recognises:

- **Single spectrum** — two columns: wavenumber, intensity.
- **Multiple spectra (columns)** — the first column is the wavenumber axis; each subsequent column is one spectrum.
- **Multiple spectra (rows)** — the first row is the wavenumber axis and each subsequent row is one spectrum.
- **Renishaw long format** — a header containing #Wave and #Intensity, optionally with #X #Y (a scan) or #Time (a series). The long table is reshaped back into a proper spectrum, line or map cube.

Note The same Renishaw long format is what BioRaman writes when you export processed maps to CSV/TXT (section 20), so processed data round-trips back into the viewer.

5 RAMANMETRIX Datasets — ZIP + Metadata

Open **File > RAMANMETRIX Dataset (ZIP + metadata)** to import a ZIP of spectra together with a metadata table (a CSV/XLS/XLSX whose filename contains the word "metadata").

5.2 Generate a metadata template

- **Set Columns** — Full default set (all twelve RAMANMETRIX columns) or Core columns (file, include, standard, date, device, batch, type).
- **Set Template** — Long (one row per file), Short (one row per \$type/\$batch/ folder), or Auto.
- Click **Generate template CSV...**, edit the CSV, then place it back inside the ZIP named metadata_*.csv for a self-describing dataset.

5.3 Apply an existing metadata table

Click **Apply metadata** to resolve each spectrum's labels. Matching follows the RAMANMETRIX rules: a row whose file/path value is a prefix of a spectrum's path applies to it; when several match, the longest pattern wins; include = False drops a spectrum; the patterns * and . match every file not otherwise listed.

5.4 Load a single spectrum from the archive

Select any file row and click **Load selected spectrum** to extract it and open it in the main map viewer, preprocessed with your current settings.

6 Preprocessing

Preprocessing is applied automatically when a file is loaded, using the settings in **Preprocessing › Settings**. The pipeline runs in this order: cosmic-ray removal, dark/pedestal subtraction, baseline correction, Savitzky–Golay smoothing and normalisation.

Step	Option	What it does
Cosmic-ray removal	on / threshold / width	Detects and interpolates over sharp spikes using a modified Z-score on the first derivative.
Dark subtraction	on / off	Subtracts the per-spectrum minimum (pedestal).
Baseline	asls / arpls / drpls / none	Whittaker baseline removal of broad fluorescence background (needs pybaselines).
Smoothing	Savitzky–Golay window / poly	Reduces noise while preserving peak shape.
Normalisation	max / area / snv / vector / none	max scales to the tallest peak; area to integrated intensity; SNV / vector remove multiplicative scaling; none keeps raw counts.

Click **Apply & Close** to save the settings. To see exactly what was applied, open **Preprocessing › Processing Log**.

Note Preprocessing choices change analytical output. Save a recipe (section 7) so your maps and statistics are reproducible.

7 Preprocessing Recipes & Reprocessing

A **recipe** is a JSON file holding every preprocessing parameter. Recipes make an analysis reproducible and easy to share, and they drive batch and command-line processing.

- **Preprocessing › Save Recipe...** writes the current settings to a .json file.
- **Preprocessing › Load Recipe...** reads a recipe back in and offers to re-apply it to the loaded data immediately.
- **Preprocessing › Reprocess** re-runs the pipeline on the retained raw cube using the current settings — no need to reload the file from disk.

Note Because the raw cube is kept in memory for the file loaded in the current session, you can tune the recipe and Reprocess repeatedly to compare results quickly. The raw data is not retained across application restarts.

8 Viewing & Navigating Spectra

- **Pick a pixel:** click anywhere on the map; its spectrum appears on the right.
- **Compare spectra:** right-click the map to add the clicked spectrum to a comparison overlay. Use Clear comparison to reset.
- **Peak markers:** toggle View > Show peak markers to annotate detected peaks.
- **Normalise view:** toggle View > Normalise spectrum to compare peak shapes independent of absolute intensity.
- **Save the current spectrum:** File > Save Spectrum (Ctrl+S).

9 Building Maps

Maps turn the spectral data cube into a single colour image where each pixel's colour encodes a chemical quantity.

9.1 Univariate Analysis

Build a map from a single spectral feature — peak intensity, integrated band area, or intensity at a single point. Set the band position and limits in cm^{-1} .

9.2 Ratio and RGB maps

Use the MODE selector to show a band-ratio map or render two bands as colour channels (A+B RGB).

9.3 Curve-Fit Map

The Curve-Fit Map window fits a peak model to every pixel and maps a fitted parameter (position, width or amplitude). Set minimum peak width and height limits to avoid degenerate fits, then click Run Fit & Create Map. Fitting every pixel is computationally heavy and may take time on large maps.

10 Regions of Interest (ROI) & ROI Analysis

An ROI restricts analysis to a chosen area. On the main map use **Draw ROI**, choose a shape (rectangle, ellipse, polygon or freehand), and draw on the map. Tick **Mask outside ROI** to hide everything beyond the region.

Click **Analyse ROI** for a guided workflow: define bands → binarize → heatmaps → statistics, with Export Figure / Export Panels options.

11 Colour & Display Control (LUT)

The LUT (look-up table) Control window fine-tunes how intensities map to colour, without changing the data: choose the colour scheme, blend opacity over the white-light image, and set vmin / vmax intensity limits manually.

12 PCA Analysis

Open from **Analysis** › **PCA**. PCA finds the main axes of spectral variation.

Control	Default	Meaning
Components	3	Number of principal components (2–10).
Wavenumber min / max	500 / 3500	Spectral window used (cm ⁻¹).
Spectra/file (0 = all)	0	Randomly sub-sample this many spectra per file to speed up very large datasets.
Remove outliers (>2 σ)	On	Discard extreme spectra before fitting.
Standardise features	Off	Scale every channel to unit variance before PCA.

Click **Run PCA**. The figure shows scores, loading spectrum, score distribution, the explained-variance scree plot, and the spatial score map. Click Save PCA Figure to export.

Note The spatial score map needs the original X×Y geometry, so it is available only when a single map with a known pixel layout is loaded.

13 Cluster Analysis

In brief: clustering automatically groups pixels with similar spectra into a chosen number of classes, producing a colour-coded class map and a representative mean spectrum per class. Choose K-means or Agglomerative (hierarchical) clustering and the number of clusters (2–10).

Note After clustering, BioRaman reports the mean **silhouette score** (range –1 to 1; higher means better-separated clusters) in the status line, to help you choose a sensible number of clusters.

14 MCR-ALS — Multivariate Curve Resolution

In brief: MCR-ALS unmixes the data into a small number of pure component spectra and their abundance maps, using alternating least squares.

Control	Meaning
Non-negative abundances	Constrain abundances to be ≥ 0 .
Non-negative spectra	Constrain recovered spectra to be ≥ 0 .
Pixel bin (k×k)	Bin pixels before fitting to speed up ALS; full-resolution maps are reconstructed afterwards.
float32	Single precision (faster, lower memory).

Click **Run MCR-ALS**. From the results you can Compare Components, Identify Peaks, Save Figure, or Save Pure Spectra (.csv).

15 N-FINDR — Endmember Extraction

In brief: N-FINDR finds the purest spectral signatures (endmembers) actually present in the data, then computes how much of each is present at every pixel by non-negative least squares (NNLS). Set the number of endmembers and the spectral window, then Run N-FINDR. Use Compare Endmembers, Identify Peaks, Save Figure or Save Endmembers (.csv).

16 Peak Identification

The Peak Identification window detects peaks in a selected spectrum and matches them against a Raman band library, listing each peak with its position (cm^{-1}), relative intensity and a candidate chemical assignment. You can import a custom band list / CSV and export the identified-peak table.

17 Spectra Comparison

The Spectra Comparison window overlays selected spectra side by side — picked pixels, cluster components or extracted endmembers — and exports the comparison figure as PNG or PDF.

18 Spectral Tools

Open from **Analysis** › **Spectral Tools**. These utilities modify the loaded map in place.

Tool	What it does
Spectral Resampling	Resample all spectra onto a new equidistant wavenumber grid (start, end in cm^{-1} , number of points).
Spatial Crop	Crop the map to a pixel bounding box.
Map Rotation	Rotate the map by 0 / 90 / 180 / 270 degrees.
Background Subtraction	Subtract an optical substrate / background reference, with a subtraction scale from 0 to 2.

Note These operations are applied in place. Use **Preprocessing** › **Reprocess** to restore the freshly-preprocessed cube, or reload the original file.

19 3D Confocal Volume Viewer

In brief: the 3D viewer turns a depth-resolved (confocal) Raman dataset into an interactive three-dimensional rendering. Open from **Analysis** › **3D Volume Viewer**. Set the Z step (μm), then load a Z-stack of slice files (or use the current 2D map for a quick look). Modes include Volume Scatter, Orthogonal Slices, Surface and multi-band RGB. Adjust threshold, voxel alpha, Z scale, pre-smoothing, lighting and camera angles, then Render 3D View and Export PNG / PDF at 250 dpi.

Note Rendering is the most memory-intensive feature. If it is slow, raise the threshold, increase pre-smoothing, or reduce the volume size.

20 Saving Processed Data

File › Save Processed Data (Ctrl+Shift+S, or the toolbar button) exports the full preprocessed spectral cube — baseline-corrected, smoothed, normalised and cosmic-ray-cleaned — together with the wavenumber axis.

Format	Notes
.npz	Compressed NumPy archive (xdata, spectra cube, processing report). Recommended: lossless and fast to reload.
.h5 / .hdf5	HDF5 dataset (needs h5py). Portable, widely supported.
.csv / .txt / .dpt	ASCII. Maps are written in the Renishaw long format (#X #Y #Wave #Intensity) so they round-trip back through the built-in reader; single spectra are two columns.
.mat	MATLAB v5 file (needs SciPy).

Note Proprietary instrument containers (.wdf / .wip / .spc) are binary formats that cannot be rewritten, so the processed data is exported to these open formats instead.

21 Quality-Control Maps

Preprocessing › Quality Control Maps (or the QC Maps toolbar button) shows per-pixel quality metrics so you can spot bad pixels before analysis. Choose a metric from the drop-down; each map has a colour bar and a min / median / mean / max summary, and can be exported to CSV or NPY.

Metric	Meaning
Signal-to-noise ratio	Peak-to-peak signal divided by a high-frequency noise estimate.
Total intensity	Sum of the processed spectrum at each pixel.
Max intensity	Maximum processed intensity at each pixel.
Raw max (saturation)	Maximum raw intensity, normalised to the global maximum — high values may indicate detector saturation (shown when the raw cube is available).

22 Batch Processing

File › Batch Process Folder applies the current recipe to every supported file in a folder. Choose the input folder, an output folder and the output format (.npz, .h5, .csv, .txt or .mat); optionally recurse into sub-folders and load a specific recipe. Click **Run batch** — progress is logged in the window.

Each input produces one processed file named *name_processed.<ext>*, plus a **batch_summary.csv** recording the status, map size and cosmic-ray count for every file.

23 Reports & Sessions

23.1 HTML analysis report

File › Save Analysis Report (HTML) writes a single self-contained HTML file bundling the mean-intensity map, the mean spectrum (± 1 SD), the preprocessing recipe and the processing log. The images are embedded, so the report can be shared as one file.

23.2 Sessions

File › Save Session stores the source file path, the preprocessing recipe and view settings (colour map, normalisation, peak markers) in a .bioses file. **File › Load Session** restores those settings and re-opens the original file if it can still be found.

Note A session stores a reference to the data file, not the data itself. Keep the original file in place so the session can reload it.

24 Headless Command-Line Mode

BioRaman can preprocess files without opening the GUI — useful on servers or in automated pipelines. Running with no arguments launches the interface as usual; passing arguments runs in headless mode.

Argument	Meaning
--input, -i	Input file or folder of Raman files.
--out, -o	Output folder for processed data.
--recipe, -r	JSON preprocessing recipe (defaults to built-in params).
--format, -f	Output format: .npz, .h5, .csv, .txt or .mat.
--recursive	Recurse into sub-folders when --input is a folder.
--save-recipe PATH	Write the default recipe to PATH and exit.
--version	Print the version and exit.

Example — write a recipe, then batch-process a folder:

```
python bioraman.py --save-recipe recipe.json
python bioraman.py -i data/ -o processed/ -r recipe.json -f .npz --recursive
```

25 Exporting Results

Most windows offer their own save buttons. The common outputs are:

Output	Where	Format
Current map	File > Save Map (Ctrl+M)	PNG / TXT / NPY
Current spectrum	File > Save Spectrum (Ctrl+S)	TXT / CSV
Processed data cube	File > Save Processed Data (Ctrl+Shift+S)	NPZ / HDF5 / CSV / TXT / MAT
Preprocessing recipe	Preprocessing > Save Recipe	JSON
Quality-control map	QC Maps window	CSV / NPY
Analysis report	File > Save Analysis Report	HTML
Session	File > Save Session	.bioses (JSON)
Batch outputs	File > Batch Process Folder	per-file + summary CSV
Figures (PCA, MCR, 3D...)	each analysis window	PNG / PDF / CSV

26 Keyboard Shortcuts

Shortcut	Action
Ctrl+O	Open a Raman file (any supported format)
Ctrl+M	Save the current map
Ctrl+S	Save the current spectrum
Ctrl+Shift+S	Save the processed data cube
Left-click on map	Select a pixel / show its spectrum
Right-click on map	Add the clicked spectrum to the comparison overlay

27 Troubleshooting & FAQ

My .txt / .csv file would not load, or loaded with a nonsensical axis

This build ships a native text reader. If a Renishaw long-format file (with #X #Y #Wave #Intensity columns) still looks wrong, confirm the header line is present and that the wavenumber column is named #Wave.

WDF loading is disabled

The renishawWiRE package is not installed. Install it with `pip install renishawWiRE` and restart.

Opening a .wip file reports that RamanSPy is needed

WITec .wip is a proprietary binary format and needs a backend. Install it with `pip install ramanspy` (or the `photonicsdata wip_loader` package).

Saving to .h5 or .mat fails

HDF5 export needs `h5py` (`pip install h5py`); MATLAB export needs `SciPy` (`pip install scipy`). Use `.npz` if you prefer a dependency-free archive.

Baseline options are missing

The `pybaselines` package is not installed. Install it with `pip install pybaselines`.

I changed the recipe but the map did not update

Use **Preprocessing** > **Reprocess** to re-apply the current recipe to the loaded data, or reload the file. The Processing Log confirms what was applied.

Quality-control saturation map is missing

The saturation metric needs the raw cube, which is retained only for files loaded in the current session. Reload the file if necessary.

28 Glossary

Term	Meaning
Hyperspectral map	A dataset with a full spectrum recorded at every pixel.
Wavenumber (cm ⁻¹)	The x-axis of a Raman spectrum; position of spectral features.
Baseline correction	Removal of broad background (e.g. fluorescence) under the peaks.
Recipe	A JSON file holding all preprocessing parameters for reproducibility.
SNR	Signal-to-noise ratio; a per-pixel data-quality metric.
Silhouette score	A cluster-separation quality metric (range -1 to 1).
ROI	Region of Interest — a sub-area selected for analysis.
PCA	Principal Component Analysis; finds main axes of spectral variation.
MCR-ALS	Multivariate Curve Resolution; unmixes data into pure spectra and abundance maps.
Endmember	A pure spectral signature present in the data (extracted by N-FINDR).
LUT	Look-Up Table; the mapping from intensity values to display colours.

Appendix A File-format quick reference

Extension	Type	Backend / requirement
.wdf	Renishaw WiRE binary map/spectrum	renishawWiRE
.wip	WITec Project binary	ramanspy
.txt / .csv	ASCII spectra & tables; Renishaw long format	built-in
.dpt / .dat / .asc	Generic ASCII spectra	built-in
.jdx	JCAMP-DX style text	built-in
.xlsx / .xls	Tabular spectra (PCA input)	pandas + openpyxl
.zip	RAMANMETRIX dataset (spectra + metadata)	built-in (\$5)
—	Processed export: NPZ / HDF5 / CSV / TXT / MAT	numpy / h5py / scipy

Appendix B Metadata column reference

Default RAMANMETRIX metadata columns (case-sensitive). Boolean columns accept True/False.

Column	Default	Meaning
file	—	File name or path pattern the row applies to.
include	True	Set False to drop a spectrum from the dataset.
standard	False	Marks calibration-standard spectra.
date	—	Measurement date (each date calibrated separately).
device	—	Measurement device name/id.
batch	—	Data segments for batch-out cross-validation.
type	—	Default class labels — the main grouping variable.

Additional custom columns may be added as regression responses, alternative grouping variables, or aggregation keys.

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